



PRINCIPAL SOFTWARE ENGINEER

Komal Shehzadi

Frontend architecture, performance engineering & AI-enabled delivery

BASE Lahore, Pakistan
EMAIL shehzadikomal303@gmail.com
WEB dev.komalshehzadi.com
EXP 6+ years in production

OPEN TO

Principal / Staff Engineer · Frontend Architect · Lead Frontend or Full-Stack · AI-Enabled Engineering

AI & AGENTIC ENGINEERING

- LLM / GenAI APIs
- Agentic workflows
- MCP integrations
- Custom subagents
- Prompt design
- AI-assisted review
- AI-assisted planning

CORE LANGUAGES

- TypeScript
- JavaScript ES2022+
- C#
- SQL
- HTML5 / CSS3

FRONTEND

- Angular
- React
- RxJS
- Redux / Thunk
- Design systems
- Tailwind
- Material UI
- shadcn/ui

VISUALIZATION & PERFORMANCE

- WebGL
- D3.js
- Konva.js
- Canvas rendering
- Lazy loading
- State optimization

OBSERVABILITY

- Grafana Faro
- Loki / LogQL
- Telemetry tiering
- Business events

BACKEND, DATA & PLATFORM

- .NET Core
- REST APIs
- SQL Server
- Firestore
- ClickHouse
- RabbitMQ
- Docker
- Kubernetes
- AWS / Azure / GCP
- Jenkins
- Bitbucket Pipelines

PROFILE

01

Principal engineer with **6+ years shipping data-intensive products** in domains where being wrong is expensive — investment banking, securities regulation, oil & gas, semiconductor yield, enterprise supply chain. I own frontend architecture end to end: the component systems teams build on, the rendering and state work that keeps million-point interfaces fast, and the **observability that explains what actually happened in production**. Lately that ownership extends to how the team builds — **agentic tooling, purpose-built AI subagents and MCP-connected workflows** that compress triage, root-cause analysis and delivery into a single loop.

~70%

Faster rendering on million-pixel semiconductor datasets

100K+

Product units traceable in a live supply-chain UI

~50%

Cut in report-generation time, oil & gas program

4x

Achiever of the Month, plus AI Hackathon podium

EXPERIENCE

02

Dec 2025 – Present
Lahore, PK

Principal Software Engineer

Techlogix – UBP Investment Banking · SECP LEAP

UBP Investment Banking

Sep 2026 – Present

- Deliver features **end to end across frontend, backend services and data** for an investment-banking business, where accuracy and reliability are non-negotiable.
- Contribute to **delivery planning** — scoping and breaking down work, estimating effort and sequencing dependencies alongside product and engineering leads.

SECP LEAP — national securities-regulator portal

Dec 2025 – Sep 2026

- Owned **frontend architecture** for a high-stakes government regulatory platform, standardising it around a **dynamic form engine and shared component library** so new statutory workflows shipped as configuration rather than copied screens.
- Instrumented the portal with **Grafana Faro** — tiered HTTP telemetry, named business events and LogQL queries over Loki that reconstruct one applicant's journey, turning "it failed for a single user" into a reproducible timeline.
- Led **production root-cause analysis across the frontend/backend boundary**: traced recurring API failures to fail-open submit gates and mis-sequenced role APIs, then shipped verified fixes through a dual-branch staging and hotfix release process.
- Built the team's **AI-assisted delivery workflow** — purpose-built subagents for observability instrumentation and defect-to-PR automation, wired into Jira and Bitbucket through MCP tooling.

Mar 2023 – Dec 2025

Lahore, PK

- Set engineering standards for conventions, review depth and documentation while **mentoring junior engineers** and acting as the technical bridge across product, backend and QA.
- Improved responsiveness across complex regulatory workflows through **lazy loading and state-management restructuring**.

Senior Software Engineer

Techlogix – visualization, traceability & analytics products

- Architected and built a **React + D3.js + WebGL visualization engine** from scratch for million-pixel semiconductor datasets, achieving **~70% faster rendering** and making previously impractical in-browser analysis routine.
- Engineered the frontend for **Authentix Track & Trace**, supporting real-time traceability across **100,000+ product units** and cutting manual product-verification time by ~60%.
- Rebuilt data-dense operational reporting for an **oil & gas digital program**, reducing report-generation time by **~50%** and improving live operational visibility.
- Analysed **11 end-to-end YieldWerx workflows**, removed process bottlenecks for a **~25% cycle-time reduction**, and left reusable templates behind for standardised operations.
- Shipped a **React + Firestore + Redux Thunk** scheduling product from concept to production, replacing fragmented coordination with one source of truth and cutting scheduling conflicts by ~40%.
- Built an **AI-powered design-to-code generator** on GenAI APIs, compressing multi-day HTML/CSS prototyping cycles into minutes.
- Delivered an **Angular + Konva.js canvas label designer** for HP, reducing label-formatting errors by ~30% and eliminating manual rework.

Oct 2020 – Mar 2023

Lahore, PK

Software Engineer

Techlogix – Campus on Cloud

- Developed and owned key modules of a **full-stack campus management platform** spanning student lifecycle, finance and academic grading, on AngularJS/Angular, Kendo UI and .NET Core.
- Automated manual administrative reporting through robust CRUD services and streamlined UX flows, improving data accuracy and removing repetitive data entry.

Aug 2020 – Sep 2020

Lahore, PK

Software Engineer

NetSol Technologies – framework training, database schema & query work

AI-ENABLED ENGINEERING

03

DELIVERY

Full-stack delivery

On UBP's investment-banking engagement, building features end to end across the interface, services and data layer — every change reviewed and verified before it ships.

PLANNING

AI-assisted planning

Use AI to turn requirements into scoped, estimable work — breaking features into tasks, drafting acceptance criteria and surfacing dependencies and risks early.

AGENTIC TOOLING

Purpose-built engineering subagents

Authored domain-specific AI agents that own real tasks — adding observability instrumentation correctly, and carrying a defect from root cause through to a reviewed pull request — with team conventions encoded as their instructions.

TOOL INTEGRATION

MCP-connected delivery loop

Connected issue tracking and source control through MCP so triage, root-cause write-ups, branch discipline and PR drafting happen in one continuous loop instead of four context switches.

PRACTICE

Standards for AI-assisted work

Define where generated code is trusted and where it is not: build verification before every PR, confirmed root causes over plausible-sounding explanations, and review depth that scales with blast radius.

PRODUCT

GenAI design-to-code generator

Built an AI-powered HTML/CSS generator on GenAI APIs that turned multi-day design-to-code prototyping into a minutes-long cycle — third place across the engineering org at the Techlogix AI Hackathon.

EDUCATION & RECOGNITION

04

EDUCATION

MS, Computer Science

UET Lahore · 2020–2023 · GPA 3.6 / 4.0

BS, Computer Science

UET Lahore · 2016–2020 · GPA 3.6 / 4.0

RECOGNITION

Achiever of the Month ×4

Techlogix · 2021, 2022, 2023, 2025

3rd Place, AI Hackathon

Techlogix · 2024 · org-wide

CERTIFICATIONS

Front-End Web UI Frameworks & Tools

Coursera · HKUST

Node.js, Express & MongoDB Bootcamp

Udemy · 2022